Investigating How Iron Supplementation Type Influences Gut Microbiome Composition and Iron Biomarkers in 12-Month-Old Anemic Infants

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Introduction and Background

Anemia is a health condition that occurs when there is a deficiency of healthy red blood cells or hemoglobin. Red blood cells (RBCs) transport oxygen from the lungs to tissues throughout the body, while hemoglobin is an iron-rich protein within RBCs that binds to oxygen and enables its transport through the bloodstream (1). Therefore, in anemic patients whose RBC or hemoglobin is deficient, blood has reduced oxygen delivery throughout the body (2). This lack of oxygen contributes to symptoms such as fatigue, weakness, and impaired cognition that many anemic patients experience (3).

Anemia can occur due to various reasons, including low birth weight or vitamin deficiency, but iron-deficiency anemia (IDA) is the most common form (4). Infants are especially vulnerable to anemia because of their rapid growth and increased nutritional requirements during early development. Iron deficiency is therefore a major contributor to anemia during infancy because iron is an essential nutrient for hemoglobin production and the formation of red blood cells (5).

If left untreated, anemic children may develop mental or motor development issues (2). Consequently, supplements such as ferrous sulphate ($FeSO_4$) or micronutrient powders (MNPs) are often recommended

to infants to help increase iron levels (4). $FeSO_4$ is a synthetic agent used to provide iron to restore low iron levels, while MNPs are micronutrient powders containing vitamins and minerals such as vitamin A and iron that can be ingested (6, 7). Previous research has been done to observe the relationship between anemia and microbial diversity in infants, and it was found that there are no structural or functional differences in their microbiome compared to healthy infants (4, 8). However, when taking iron supplementation, infant gut microbial composition does become altered, such as reductions in Bifidobacteria and lactobacilli, both of which are beneficial bacteria (9, 10).

Although both MNP and $FeSO_4$ are common types of supplements used to treat anemic patients, their effect on microbial composition relative to each other is not known. Furthermore, although many studies were previously done to observe the change in ferritin and hemoglobin levels after using $FeSO_4$ supplementation, little research has been done on MNP supplements.

This knowledge gap is essential because although $FeSO_4$ raises ferritin and hemoglobin levels, $FeSO_4$ is not always ideal because it reduces beneficial bacteria and can cause side effects like nausea and fever due to its higher iron dosage compared to MNP (11). Understanding how different iron supplementation strategies influence the gut microbiome may help identify interventions that improve infant iron status while minimizing disruptions to beneficial microbial communities.

To investigate this knowledge gap, we aim to use the 16S rRNA sequencing data from McClorry et al. to analyze iron supplementation type (MNP vs. $FeSO_4$ vs. none) influences gut microbiome composition and how these microbial differences relate to hemoglobin and ferritin levels in 12-month-old breastfed infants. The samples in the anemia dataset from McClorry et al. were collected in Iquitos, Peru, and they involved 102 infants at 12-13 months of age that appeared healthy and were born at a healthy weight (≥2500 g) (12). These infants were then administered MNP, $FeSO_4$, or no supplements, and venous blood samples were collected from each infant to measure hemoglobin ferritin levels.

Hemoglobin and ferritin levels for each sample were then measured using these blood samples. Since measuring RBC for anemia diagnosis requires complex analysis, only hemoglobin concentrations were used to determine if an infant was anemic. If an infant had a hemoglobin concentration of <110 g/L, they were considered anemic. Samples for 6-month-old infants were also included, but we are only focusing on the 12-month-old infants for our studies since the microbiome changes quickly in early life development. For our study, we plan to focus on hemoglobin and ferritin levels, which are blood biomarkers used to assess iron status. Hemoglobin measures the concentration of the oxygen-carrying protein in RBCs and is used to identify anemia, while ferritin measures the amount of iron in the body (12). For our analysis, low hemoglobin levels are defined as having <110 g/L and normal levels as 110 g/L or higher (4, 12). For ferritin levels, low levels are defined as having <12 µg/L and normal as 12µg/L or higher (4, 13).

Hypothesis & Prediction

Among 12 month old breastfed infants, how does the type of iron supplementation (MNP vs. FeSO4 vs. none) influence gut microbiome composition, and how do these microbial differences relate to hemoglobin and ferritin levels?

As discussed above, anemia is associated with a reduction in hemoglobin concentration and can be treated through methods such as iron supplements or an increased intake of dietary iron (14). Previous studies have suggested that iron supplements can decrease the risk of low levels of ferritin and hemoglobin (15). Additionally, others suggest that iron supplementation can alter the microbiome composition, such as one study by Karamantziani et al.(16) which found a decrease in Bifidobacteria in patients who were taking ferritin-guided iron supplements. Thus, we hypothesize that 12 month old anemic infants receiving iron supplementation (MNP or FeSO4) will exhibit higher levels of ferritin and hemoglobin levels compared

to the control group, infants receiving no supplementation. In addition, we predict that infants receiving iron supplementation will display statistically significant gut microbiome diversity, taxonomic composition, and predicted microbial functions compared to the control group. These microbiome differences may reflect microbial responses to increased iron availability within the gut microbiome.

Experimental Aims & Rationale

AIM 1: Determine whether gut microbiome diversity differs across iron supplementation groups

The first aim of this study is to determine whether iron supplementation type (MNP, FeSO4 or none) is associated with changes in gut microbiome diversity in 12-month-old anemic breastfed infants. Microbial diversity is widely considered an indicator of microbiome stability and ecosystem health. Changes in alpha and beta diversity within the gut community can reflect dysbiosis or shifts in the ecological structure of microbial populations. Important changes to the makeup and resilience of one's microbial community may be affected through nutritional interventions during infancy which is a crucial period for the development of the microbiome (17). Previous research has shown that anemia and iron deficiency in infants are associated with altered gut microbiome composition and microbial metabolic function, supporting the idea that iron status may influence the developing gut microbiome (4).

To help determine whether different iron interventions are associated with broader microbial differences in the gut microbiome, microbial diversity will be compared across anemic infants receiving MNP, FeSO4, or no supplementation. Alpha diversity will be assessed to evaluate within-sample microbial diversity, using metrics such as Shannon diversity, which accounts for both richness and evenness, and

observed features which capture total taxonomic richness. These metrics provide insight into how complex and balanced microbial communities are within individual infants. In contrast, beta diversity will be assessed to evaluate differences in overall microbial community composition between samples, allowing us to determine whether microbial communities cluster differently across supplementation groups. Together, these diversity metrics allow us to determine whether iron supplementation alters the overall ecological structure of the infant gut microbiome before examining specific taxa or functional differences.

Examining these diversity patterns is biologically important because they may reveal whether iron supplementation influences microbiome maturation, microbial stability, and the balance of microbial communities during a sensitive developmental period in anemic infants (4, 17). Establishing whether such overall differences exist is an essential first step in addressing our research question, as it determines whether supplementation groups differ at the community level before examining specific taxa or potential associations with hemoglobin and ferritin levels.

AIM 2: Identify microbial taxa that differ in abundance across iron supplementation groups

While Aim 1 examines overall community-level differences in the microbiome, further analysis is required to determine which specific microbial taxa are responsible for these observed differences.

Although diversity metrics provide insight into overall microbial community structure, they do not identify which specific microbial taxa are responsible for differences between groups. Therefore, the study's second aim is to identify microbial taxa whose relative abundance differs across the infants

receiving MNP, FeSO4 and no supplementation. By comparing these supplementation groups, this aim will determine which microbes are most strongly associated with each supplementation.

This aim is biologically relevant because iron is an essential nutrient not only for the human host but also for many gut microbes. Thus altering iron availability may reshape microbial competition within the gut microbiome (18, 19). Microorganisms that possess efficient iron acquisition mechanisms may gain a selective advantage when iron levels increase, potentially altering the balance of microbial communities (18, 19). Measurable shifts in the taxonomic composition of the infant gut microbiome may be seen through different supplementation groups favoring the growth of distinct microbial taxa.

Previous studies have shown that less favorable taxonomic profiles are observed in infants as the gut microbiome was observed to be altered through the administration of different iron-containing supplements. For example, iron supplementation has been associated with reductions in beneficial commensal bacteria such as Bifidobacterium and increases in potentially pathogenic taxa, including members of the Clostridium genus and other opportunistic microbes (19, 20).

Through the identification of differing taxa across these supplementation groups, which include comparisons between MNP/control, FeSO4/control, and MNP/FeSO, this aim will reveal which microbial populations are most responsive to which supplementation group. Through these findings, the relation between the three supplementation groups and their influence on the taxonomic composition of anemic infant gut microbiomes will be better understood. Furthermore, taxa identified as significantly associated with supplementation groups may represent microbial indicators of iron exposure and will be further evaluated in Aim 3 to determine whether these microbial patterns are linked to hemoglobin and ferritin levels. Identifying such taxa is therefore a crucial step toward understanding how iron supplementation may influence both microbiome composition and iron-related health outcomes such as inflammation or

dysbiosis, in infants.

AIM 3: Determine whether predicted microbial functional pathways differ across supplementation groups or vary alongside iron biomarkers

After identifying specific taxa associated with supplementation groups in Aim 2, it is important to determine whether these microbial shifts also correspond to changes in microbial metabolic potential..

Taxonomic composition alone does not fully explain how the microbiome may influence host biology. Therefore, the third aim of this study is to investigate whether predicted microbial functional pathways differ across supplementation groups or are associated with hemoglobin and ferritin levels. Previous taxonomic analyses are being built on by this aim which examines whether microbial communities associated with different iron supplementation strategies also differ in their predicted metabolic capabilities. Fermentation, vitamin synthesis, immune modulation, and nutrient metabolism are some of the numerous metabolic processes relevant to host physiology that gut microbes contribute to. The microbial community's functional potential may also be altered via shifts in microbial composition (4, 10, 18).

Anemia and iron deficiency not only is associated with changes in microbial composition, but with alterations in metabolic pathways and metabolite production previous research suggests. Taxonomic shifts being linked to microbial functional differences are indicated through anemic infants exhibiting changes in microbial metabolic activity and systemic metabolite profiles (4). These findings suggest that iron availability may influence the metabolic roles that microbes play within the gut microbiome.

By examining predicted microbial pathways across infants receiving MNP, FeSO4 or no supplementation,

this aim will determine whether different iron interventions are associated with functional differences in the gut microbiome. If specific metabolic pathways differ across supplementation groups, this would suggest that iron supplementation may influence not only which microbes are present but also the biological activities those microbes perform within the infant gut. In addition, evaluating whether predicted pathways vary alongside hemoglobin and ferritin levels may reveal microbial metabolic patterns that track with iron status.

Together, these findings may provide insight into potential mechanisms linking the gut microbiome with iron metabolism in infants. Identifying functional pathways that differ across supplementation groups or correlate with iron biomarkers will generate hypotheses about how microbial metabolic activity may influence host iron status. These functional insights will complement the taxonomic findings from Aim 2 and contribute to a more comprehensive understanding of how iron supplementation may shape both the composition and metabolic potential of the infant gut microbiome.

AIM 4: Assess whether microbiome diversity metrics or differentially abundant taxa are associated with hemoglobin and ferritin levels

It is vital to determine whether these clinically relevant indicators of iron status are associated with these microbiome features in order to go beyond just identifying microbial differences between supplementation groups.

The fourth and final aim of this study is to determine whether microbiome features are associated with hemoglobin and ferritin levels. While the previous aims investigate how iron supplementation influences microbial diversity, taxonomic composition, and predicted microbial functions, this aim examines

whether these microbiome features vary alongside physiological indicators of iron status. Hemoglobin is a key indicator of oxygen-carrying capacity and anemia, while ferritin reflects iron storage and is commonly used to assess iron deficiency in infants and children (26, 27). Examining the relationship between iron biomarkers and microbiome features may uncover microbial patterns that are not solely due to supplementation but linked to host physiology as well, as these biomarkers capture clinically meaningful aspects of iron status.

In particular, this aim will evaluate whether microbial diversity metrics identified in Aim 1 and microbial taxa identified in Aim 2 are associated with hemoglobin and ferritin levels. Identifying such relationships is biologically important because the gut microbiome may influence iron metabolism through several mechanisms, including microbial competition for luminal iron, modulation of intestinal inflammation, and production of metabolites that affect nutrient absorption and host metabolic regulation (10, 18). As a result, variations in microbial diversity or the abundance of specific taxa may coincide with differences in iron status among infants.

If microbial diversity measures or specific taxa vary alongside hemoglobin or ferritin levels, these associations may suggest that particular microbial communities are linked to improved or impaired iron status. Hypotheses for future mechanistic studies that investigate microbiome-host interactions in infant iron deficiency could be generated from these correlations, that would not establish correlation, but they may identify microbial patterns that track with clinically relevant iron outcomes. By linking microbial characteristics to iron biomarkers, this aim helps address the overall research question by determining whether microbiome differences associated with supplementation are also related to physiologically meaningful indicators of iron health.

A comprehensive framework is provided through these aims, in order to determine whether iron supplementation influences the infant gut microbiome at multiple levels, including microbial diversity,

taxonomic composition, metabolic potential, and associations with host iron biomarkers.

Proposed Approach

Data Processing	
A	Import anemia_metadata into the class server.
B	Import raw sequences and generate anemia_demux_seqs.qza.
C	Denoise and cluster the dataset using DADA2 to generate anemia_table.qza and anemia_rep-seqs.qza.
D	Trim and truncate dataset to 252 ntds long and generate ref-seqs-trimmed.qza.
E	Retrain classifier with req-seqs-trimmed file then generate anemia_classifier.qza.
F	Apply the newly trained classifier to the dataset to generate anemia_tax.qza.
G	Filter anemia_table.qza to remove mitochondrial and chloroplast genetic information, generating anemia-feature-frequency-filtered-table.qza and anemia-feature-frequency-filtered-table.qzv.
H	Use anemia-feature-frequency-filtered-table.qzv to find maximum sequencing depth.
I	Generate a phylogenetic tree.
J	Using anemia_metadata, generate a rarefaction curve and perform alpha rarefaction.

Aim Specific Approaches			
Aim 1: To determine how gut microbiome diversity differs across iron supplementation groups (MNP, FeSO4, and no treatment)	**Aim 2: To identify which microbial taxa whose relative abundance differs between the iron supplementation groups**	**Aim 3: To determine whether predicted functional pathways differ across the iron supplementation groups and/or are statistically associated with the iron biomarkers**	**Aim 4: To identify if microbiome diversity metrics or different taxa are associated with iron biomarkers (Hemoglobin and ferritin levels)**
Within R, install phyloseq, tidyverse, vegan, microbiome and ggplot2 (21–25, 28, 29).	Within R, install phyloseq, tidyverse, vegan, microbiome, ggVennDiagram, ggplot2, ANCOMBC, and indicspecies. (30–32).	In QIIME2, use table.qza and rep-seqs.qza as inputs for functional prediction.	Within R, install phyloseq, tidyverse, ggplot2, broom and purrr (33, 34).
Generate Phyloseq object.		Run PICRUSt2 on the feature table to predict functional pathways based on 16S data. Export results as a .tsv file (35).	Using the results from the alpha diversity analysis (Shannon index) from Aim 1, fit linear regression models for ferritin and

Aim Specific Approaches			
			hemoglobin levels.
Subset the phyloseq object into the desired groups using *group_by()* and *filter()*.		Load tidyverse, phyloseq, pheatmap and ggpicrust2 in R. Import predicted functional table using read_delim (36).	Using the taxa identified as significant in Aim 2, fit linear regression models for ferritin and hemoglobin levels.
Using the subsetted phyloseq object, perform alpha diversity analysis using the Shannon index and Observed features metrics.	Perform a core microbiome assessment to determine unique and shared taxa across iron supplementation groups.	Use *group_by()* and *filter()* to subset data into 3 groups.	Run stratified analyses by supplementation group if necessary to evaluate whether associations differ across groups.
Using the subsetted phyloseq object, perform beta diversity analysis using the Bray-Curtis dissimilarity and Unweighted UniFrac	Visualize the core microbiome using ggVennDiagram.	Perform pathway differential abundance analysis using Maaslin2 (or ANCOMBC).	Perform Spearman correlation test to assess relationships between microbiome and iron biomarkers.

Aim Specific Approaches			
metrics.			
Using *shapiro.test()*, confirm the data is non-parametric, then run Kruskal-Wallis and PERMANOVA tests to assess any significant differences between the supplements.	Analyze differences in relative abundance of core microbiome species using ANCOM-BC.	Annotate pathways and filter to only include significant pathways.	Generate scatterplots with regression lines and create summary table summarizing regression coefficients, p-values.
Generate box plots using ggplot2 for each supplementation group to visualize possible differences in diversity.	Select for taxa that have an adjusted p-value < 0.05 and pass the stress test, resulting in only taxa that are considered significant.	Perform Spearman test to assess the correlation between the metabolic pathways and iron biomarkers.	
	Visualize differences in abundance using ggplot and function *geom_col()*.	Generate bar plot using ggplot2 and a clustered heat map using heatmap to visualize functional pathway differences.	

Participation Report

Teammate	Proposal Contribution	Analysis
Aidan Chance	References & Editing	PICRUSt2
Bianca Hufana	Hypothesis/Prediction	Differential Abundance Test
Jacqueline Chan	Introduction & Background	Linear Regression Models
Rajbir Sidhu	Experimental Aims & Rationale	Alpha & Beta Diversity
Tiffany Huang	Proposed Approaches	Differential Abundance Test

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